

# Zilong Zhao | Robotics Controls Engineer

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## Education

### University of Alberta

Master of Science in Mechanic Engineering

Edmonton, AB

2024.05–2026.07(expected)

### Xi'an-Jiaotong Liverpool University

Bachelor of Engineering with Honours in Intelligent Robotics Engineering

Suzhou, China

2019.09–2023.07

Course, Robot Modeling and Control: Derivation and implementation of forward and inverse Kinematics and PD Control

## Work Experience

### BOZHON Precision Industry Technology Co.,Ltd.

Robotics Engineering Intern

Suzhou, China

2022.01–2022.08

### Xi'an-Jiaotong Liverpool University

Research Assistant

Suzhou, China

2023.07–2024.05

## Qualified Projects — More on my web page

### Metal 3D Printing Defect Repair System

Integration · Python · Testing&Validation · Customer Facing · AI

- Developed a pipeline from vision and segmentation (auto / semi-auto / manual) to path generation and ABB robot execution, integrating a 2D camera, distance sensor, and robot arm into a single system.
- Deeply customized a Python-based 'LabelMe' tool into a full-featured GUI application enabling defect labeling, boundary adjustment, parameter configuration, automatic toolpath generation, and one-click robot execution.
- Designed and ran camera-robot calibration experiments; achieved toolpath execution accuracy better than 1.5 mm on the X-axis and 3 mm on the Y-axis.
- Led on-site factory assembly, integration, and debugging; system was successfully commissioned and executed on real production parts.
- Integrated the Segment Anything Model (SAM) to enable single-click defect segmentation, significantly reducing manual effort.

### Visual-Tactile Sensor Grasping

Integration · ROS · Python · AI

- Developed an end-to-end grasping pipeline from perception and 6D pose estimation to grasp execution and force adjustment, integrating a 2D camera, vision-based tactile sensor, gripper, and UR5 robot arm into a single system.
- Designed a ROS-based architecture for data flow management (image streams, target 6D poses, control commands, robot states) and used RViz extensively for debugging and visualization.
- Trained and Inferred DL models via PyTorch. Controlled the UR5 via URX and implemented ROS nodes in Python for tf transformations, topic publishers/subscribers, and low-level motion control.
- Trained a VGG19-based model for object localization and a lightweight MLP for high-frequency tactile signal classification; built a data collection and domain randomization pipeline to improve model robustness.

### Delivery Robot Infrared AprilTag Localization

Integration · Test&Validation · Documentation · ROS

- Developed the infrared-based localization subsystem for an indoor delivery robot, integrating an IR camera, active IR emitter, and polarizing filter to detect reflective AprilTag markers and work alongside the existing SLAM system.
- Designed and executed repeatability experiments to evaluate localization accuracy under different conditions and environments.
- Documented key findings to the team: ambient light improves localization accuracy; obstacle-avoidance maneuvers have minimal impact on final pose error; tag mounting height significantly affects performance; overly dense tag placement can degrade robustness.
- Managed ROS topics and tf transforms for the localization pipeline, and recorded rosbag datasets for backup, and future experiments.

### Human Legs Tracking with 2D LiDAR

Technical Discussions · ROS · C++ · ML

- Collaborated closely with my supervisor and teammates on technical topics including Random Forest feature design and selection, Kalman filtering, and motion-prediction-based tracking.
- Recorded LiDAR rosbags for training the ML models; used RViz for visualization, debugging, and evaluation of model performance.
- Modified CMakeLists.txt and package.xml, and adapted ML code to support a higher-density LiDAR sensor.
- Implemented and optimized a Random Forest-based classifier by redefining input features and tuning model hyperparameters to improve detection accuracy and robustness.

### SO-ARM101 in IsaacSim Control by MoveIt

ROS2 MoveIt&URDF

- Generated MoveIt configuration files from the URDF using the MoveIt Setup Assistant, then performed motion planning in ROS2 and sent joint command trajectories via ROS topics to control the SO-ARM101 robot arm in NVIDIA IsaacSim.

### SO-ARM101 in IsaacLab Lift Cube

Control · Python · RL

- Integrated low-level motion and gripper control by mapping policy outputs to joint commands.
- Implemented a IsaacLab environment in Python for, defining environments, commands, observations, actions, rewards, and terminations.
- Trained a PPO policy to control the gripper to reach, grasp, and lift a cube, using stepwise reward shaping (reach → grasp → lift → reach target) to stabilize learning.

## Volunteer

### Mission Community Services Society

Meals on Wheels Driver

Mission, BC

2025.06–2026.09

### CityReach

FFF Warehouse Volunteer

Vancouver, BC

2025.06–now